Physical Changes	Chemical Changes

In class six, we read that matter is classified as solid, liquid and gas based on the physical state. We know that matter is made up of tiny particles, atoms and molecules; particles are in constant and random movement. Let us have a look at the summary of the characteristics of solid, liquid and gas.

When the arrangement of the particles in a substance change for any reason (applying pressure, altering temperature and other different reasons) the physical state of the substance gets changed.

Let us see what happens when we apply heat to the substances.

3.1 Effect of heat on solid, liquid and gases

Q17DZ

Upon heating, particle arrangement within the state of matter gets disturbed. The disturbance is seen either as expansion or contraction. When heated or cooled, the object may expand or contract, but the mass remains the same. That is, the number of particles that was inside the object does not undergo any change, only the arrangement of the particle changes. When a glass of water is heated, its volume increases and if a glass of water is cooled its volume decreases.



In which particles are very close together.

Particles are arranged in a fixed regular pattern.

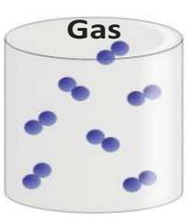
Particles can vibrate about their fixed positions.



In which particles are close together.

Particles are not arranged in a fixed regular pattern.

Particles are able to slide past one another.



Particles are far apart from each other.

Particles are not arranged in a fixed regular pattern.

Particles move freely over long distances.

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